

MediFlow-CA — Iteration 0: Problem Definition & Design (v0.0)

1. Problem Statement & Clinical Context

Field Observations

As a frontline medical interpreter, my role places me directly within communication between patients and healthcare providers. Through firsthand experience in healthcare interpreting encounters, I repeatedly observed gaps in how clinical information is transferred, accessed, and reused across healthcare settings.

These observations led me to explore a broader question:

What happens when clinical information exists, but cannot reliably move between healthcare providers and systems?

Core Problem: Fragmented Clinical Information Across Healthcare Systems

Three Real-World Perspectives

Level	Real-World Clinical Situation	What I Observed	What It Reveals
Interpersonal / Information Exchange	A hospital contacted a patient's family before an urgent biopsy to confirm whether the patient was taking blood thinners (anticoagulants). The family did not understand the medical terminology, and the hospital simply told them: "Then you'll need to confirm it with the doctor yourself." The family was frustrated: "Why are we being asked to check this ourselves? Why are we being used as the messenger?"	The patient and family were placed in the position of relaying clinical information between providers, despite having difficulty understanding the information themselves.	Clinical information cannot reliably move between providers.
Operational / Clinical Workflow	A patient visited Clinic A one month ago, Clinic B two weeks ago, and Clinic C one week ago. This week, the patient returned to Clinic A. At Clinic A, the patient was asked to confirm whether their medical conditions, medications, vaccinations, and other information were up to date, and whether anything had changed since their previous visit.	The patient had to repeatedly provide information that had already been collected. Although repeated verification can be clinically appropriate for safety, relying on patients to reconstruct existing	Existing clinical information cannot be readily accessed and reused.

	The patient felt frustrated by having to repeatedly provide the same information at each visit, describing it as “like being interrogated as if they had done something wrong,” which also took up their time. They also reported difficulty remembering medication names, dosages, and other details accurately.	information adds administrative burden and may introduce data quality risks.	
Cross-System / Cross-Border	An Asian patient who has been living long-term in North America was diagnosed with a medical condition while in Asia. Later, the patient travelled to North America to seek a second opinion. However, the patient had to carry previous medical documentation, medical images, photos, and other records from Asia to the receiving healthcare provider. The patient expressed frustration: “We don’t even speak the same language. It’s really difficult to see a doctor like this.”	The patient had to reconstruct and explain their previous medical history because the receiving provider did not have direct access to information recorded in another healthcare system.	Clinical information is not easily portable across systems.

2. Goal & Objectives

Goal

MediFlow-CA is a healthcare interoperability engine that explores how clinical information can be transformed from legacy HL7 v2 messages into standardized FHIR R4 resources, making information easier to exchange and reuse across healthcare systems.

The project explores how clinical information can be moved between systems through a process that protects sensitive information and keeps a record of how the data was processed.

Objectives

The project focuses on three areas:

- **Patient Experience:** Reduce the burden on patients and families to repeatedly reconstruct or relay clinical information.
- **Clinical Workflow & Data Quality:** Reduce unnecessary manual re-entry and the data quality risks associated with fragmented information.
- **Interoperability & Continuity of Care:** Make clinical information more portable and reusable across compatible systems through standardized data exchange.

Project Boundary

The initial design focuses on HL7 v2 → FHIR R4 interoperability for structured clinical data. MediFlow-CA is not intended to replace existing EHR systems or provide a complete healthcare information exchange infrastructure. Future iterations may explore unstructured clinical text and imaging/binary data interoperability.

3. Conceptual Architecture & Data Flow

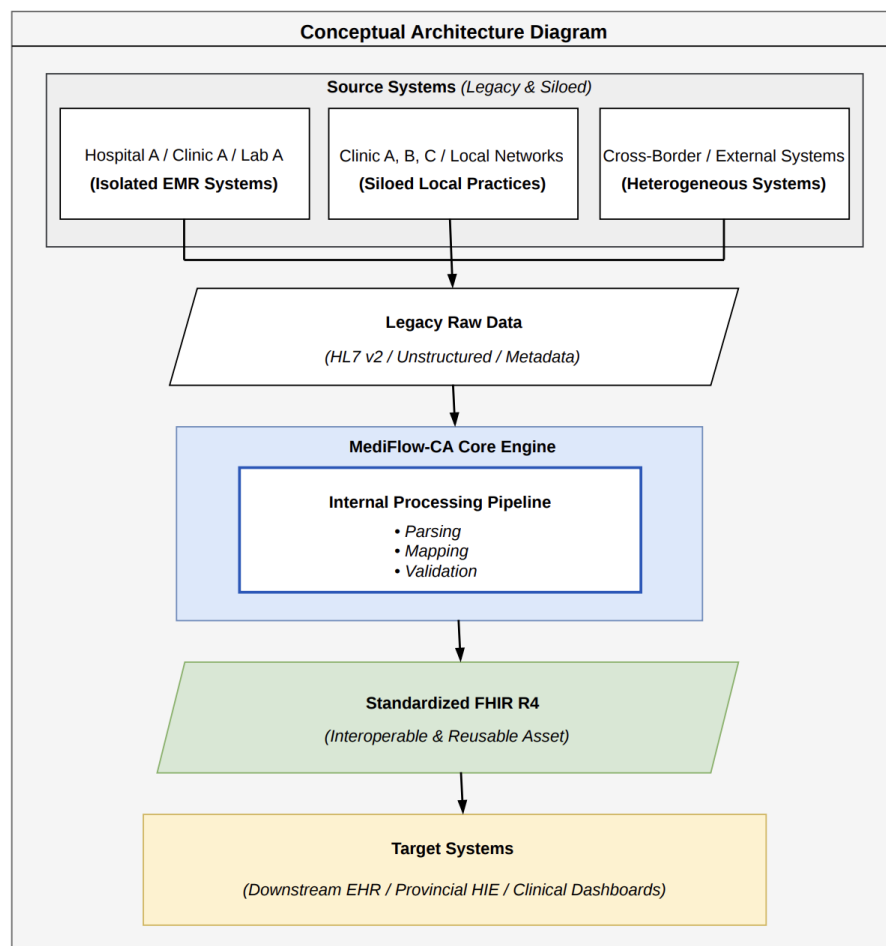


Figure 1: MediFlow-CA Conceptual Architecture & Data Flow